Unit III

1. Discuss any three Noise Probability Density Functions.
2. Explain different noise models in digital images. Discuss how Gaussian noise, salt and pepper noise and Uniform noise affect image quality.
3. Explain image restoration techniques. Discuss restoration in the presence of noise with suitable examples.
4. Explain inverse filtering method for image restoration. Discuss its advantages and limitations.
5. Explain inverse filtering and how it is used for image restoration.
6. Apply Inverse filtering technique to explain how a blurred image can be restored. Describe the steps involved and mention its limitations.
7. Write a note on any three morphological operations.
8. Explain morphological image processing. Discuss any two basic morphological algorithms with suitable examples.
9. Describe the basic morphological operations in image processing. Explain Erosion, Dilation, opening and Closing with suitable illustrations.
10. Discuss point and line detection methods in image segmentation.

    or

11. Explain the following.

    i) Detection of isolated points
    ii) Detection of line.

Unit – IV

1. Explain Pattern Recognition and discuss its fundamental concepts.
2. Explain the concept of Pattern recognition. Discuss the differences between supervised and unsupervised learning with suitable examples.

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3. Explain the fundamentals of pattern recognition. Differentiate between supervised and unsupervised learning.
4. Apply the concept of distance measures in clustering. Explain how Euclidean distance and Manhattan distance are used to group similar datapoints.
5. Discuss in brief the concept of Learning and Adaptation in Pattern recognition.
6. Explain the concept of decision boundaries and feature space in pattern recognition.
7. Explain Bayes Decision Theory and its application in Pattern Recognition.
8. Describe Bay’ s decision theory. Explain Prior probability, likelihood, posterior probability, and decision boundary.

9. Apply the concept of decision boundaries to explain how a classifier separates two classes in feature space. Illustrate your answer with a simple example.
10. List and explain several important applications of pattern recognition.
11. Explain Linear Discriminant Analysis (LDA) with mathematical formulation and applications.
12. Discuss in brief the concept of Learning and Adaptation in the context of pattern recognition

Unit – V

1. Explain Principal Component Analysis (PCA) for dimensionality reduction with suitable examples.
2. Describe Principal Component Analysis (PCA). Explain hoe PCA helps in dimensionality reduction for image classification.
3. Explain Support Vector Machines (SVM) and K- Nearest Neighbors (KNN) algorithms used in image classification.
4. Apply the concept Support vector Machine (SVM) to explain how it separates two classes using a margin hyperplane. Illustrate your explanation with a simple diagram.
5. What is the K-Nearest Neighbors (KNN) algorithm, and how is it applied in image classification?
6. Explain K- Nearest Neighbors (KNN) classifier. Describe its working procedure and mention its advantages and limitations.
7. Discuss distance functions and similarity measures used in clustering.
8. Explain Hierarchical clustering and DBSCAN clustering algorithms with suitable diagrams and applications.
9. Distinguish between training dataset and testing dataset used in pattern recognition systems
10. Compare and contrast Single layer perceptron and Multi-layer Perceptron.
11. Differentiate between normalization and standardization techniques used in data pre-processing.
12. Apply K-Means clustering algorithm to explain how data points are grouped into clusters. Describe the steps involved in forming clusters and updating centroids.